Sorting algorithms: bubble sort and insertion sort

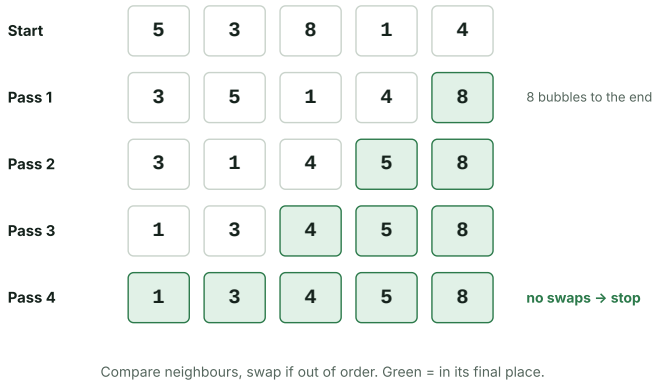
D11 FACT FILE

Specification: students should be able to

- Create and evaluate algorithms, including those for basic sorting
- Simple sorting techniques such as the bubble sort and the insertion sort

CCEA fact file: Programming Constructs 2

Key words: sort, ascending, descending, bubble sort, pass, swap, insertion sort, sorted sub-list, efficiency



Bubble sort passes on 5, 3, 8, 1, 4

Bubble sort

A bubble sort makes repeated passes through the list comparing adjacent pairs and swapping those in the wrong order. After each pass the largest unsorted item reaches its final position at the end. The sort stops when a pass completes with no swaps (or after n − 1 passes). It is easy to write but slow for large lists.

Insertion sort

An insertion sort builds a sorted sub-list at the start. Each new item is taken from the unsorted part and inserted into the correct position in the sorted part, with larger items shifted along. It is generally more efficient than a bubble sort, especially when the data is already nearly in order.

T Key terms from the mark schemes

Bubble sort
Compares adjacent items and swaps them if out of order, repeating passes until no swaps occur.
Credited: U4 2023 Q9, 2024 Q7, 2025 Q1

Insertion sort
Builds a sorted list by inserting each item into its correct position.
Credited: U4 2022 Q5

C C# example

Bubble sort in C#

```
int[] items = { 5, 3, 8, 1, 4 };
bool swapped = true;
while (swapped)
{
    swapped = false;
    for (int i = 0; i < items.Length - 1; i++)
    {
        if (items[i] > items[i + 1])
        {
            int temp = items[i];
            items[i] = items[i + 1];
            items[i + 1] = temp;
            swapped = true;
        }
    }
}
Console.WriteLine(string.Join(", ", items));
```

Exam focus

Recognise how a bubble sort works (2025 Q1(e)), tick true statements and show passes (2024 Q7(d)), order the steps (2023 Q9(c)), show passes on five values (2019 Q12(b)(ii)), and the 2022 QWC on both sorts (Q5). The 2025 report noted pupils need more practical experience of both sorts.

Unit 4 2025 Q1	[5]	String type, compound AND, IF output, NOT/AND table, bubble sort
Unit 4 2024 Q7	[35]	Arrays, pseudocode, FOR/WHILE totals, minimum, bubble sort, linear and binary search
Unit 4 2023 Q9	[18]	Arrays: update, reset to zero, find maximum; bubble sort steps; searching sorted data
Unit 4 2022 Q5	[6]	Bubble and insertion sort, efficiency (QWC)
Unit 4 2019 Q12	[8]	Linear vs binary search; showing sort passes

? Check yourself

1. Describe how a bubble sort orders a list into ascending order. [4]
2. Show the list after each pass of a bubble sort on 7, 2, 9, 4, 1. [4]
3. Describe how an insertion sort works. [3]

Answers: 1. Any four from: compare the first two items (adjacent pair) (1) / if they are in the wrong order (first larger than second), swap them (1) / move along one position and compare the next pair, repeating to the end of the list – one pass (1) / after each pass the largest remaining item has 'bubbled' to its correct place at the end (1) / repeat passes until a whole pass makes no swaps (or n − 1 passes) (1) 2. Pass 1: 2, 7, 4, 1, 9 (1) / Pass 2: 2, 4, 1, 7, 9 (1) / Pass 3: 2, 1, 4, 7, 9 (1) / Pass 4: 1, 2, 4, 7, 9 (1). (A fifth pass makes no swaps, confirming the list is sorted – not required.) 3. Any three from: the first item is treated as a sorted sub-list (1) / the next item is taken from the unsorted part (1) / and inserted into its correct position in the sorted part (1) / larger items are shifted one place right to make room (1) / repeat until every item has been inserted (1)